Ryotaro Okabe

Email: rokabe [at] mit.edu | Website | Github | Google Scholar | LinkedIn | ORCID

EDUCATION

Ph.D. in Chemistry, Massachusetts Institute of Technology, *MA, USA* / *Sept. 2021 – Sept. 2026*

M.S. in Engineering, Institute of Science Tokyo, *Japan* / *Sept. 2019–Sept. 2021*

B.S. in Engineering, Institute of Science Tokyo, *Japan* / *Apr. 2016–Sept. 2019 (Early Graduation)*

- **Courses**: Quantum mechanics, Statistical mechanics, Solid state physics, Machine learning, Group theory
- **Teaching**: Quantum Mechanics (2021), General Chemistry (2022)
- **Leadership:** APS Group on Data Science Student Representative (2026-2028), Japanese Assoc. of MIT, Treasurer (2022-2024); Sidney Pacific Graduate Residence, Officer (2022-2024); Japan-US Science Forum in Boston, Organizer (2022-2024)

EXPERIENCES

Argonne Scholar (Maria Goeppert Mayer Fellow), Argonne National Laboratory**,** X-ray Science Division
(Sponsors: Dr. Haidan Wen, Dr. Mathew Cherukara), *IL, USA* | *Oct. 2026–Present*

- Develop AI/agent-based methods for the analysis and design of X-ray measurements at the Advanced Photon Source.

Research Assistant, Massachusetts Institute of Technology (Advisor: Prof. Mingda Li), *MA, USA* | *Nov. 2021– Sep. 2026*

- Developed machine learning methods to explore the structure-property relationship in materials (e.g., prediction of phonon dispersion from material structures, generating quantum material candidates).
- Developed machine learning methods to localize the radiation sources with detector arrays.

Research Intern, IBM Thomas J. Watson Research Center (Mentor: Dr. Ching-Tzu Chen), *NY, USA* | *May– Aug. 2024*

- High-throughput calculation of electrical conductance in topological interconnect materials.
- Developed machine learning methods to predict the conductance of slab materials.

Research Assistant, Institute of Science Tokyo (Advisor: Prof. Yoshitaka Ishii), *Japan* | *Oct. 2018–Aug. 2021*

- Implemented new schemes of solid-state nuclear magnetic resonance (SSNMR) for selective signal acquisition and evaluated them through multidimensional SSNMR experiments with numerical simulations.

Research Intern, Yale University (Advisor: Prof. Charles A. Schmuttenmaer), *CT, USA* | *Jan.–Apr. 2019*

- Synthesized metal-organic frameworks (MOFs) as photosensitizers and characterized them using powder X-ray diffraction, electrochemical experiments, and time-resolved terahertz spectroscopy.

Research Intern, Rice University (Advisor: Prof. Angel A. Martí), *TX, USA* | *Aug.–Sept. 2018*

- Observed the aggregation kinetics of Amyloid beta (Aβ) protein and interactions with the metal complex.

HONORS/AWARDS

- Maria Goeppert Mayer Fellowship, Argonne National Laboratory | *Oct. 2026 – Oct. 2029*
- MRS2025S Graduate Student Silver Award | *Apr. 2025*
- MRS2025S Best Poster Award | *Apr. 2025*
- IBM PhD Fellowship Awards (Granted 40,000 USD) | *Sept. 2024 – Aug. 2025*
- NERSC GenAI AY2024 Resource Allocations (6,800 GPU node hours, 1,000 CPU node hours) | *Jul. 2024*
- JASSO Graduate Fellowship (Granted approx. 11,000 USD) | *Sept. 2023 – Aug. 2024*
- Heiwa Nakajima Foundation Fellowship (Granted approx. 70,000 USD) | *Sept. 2021 – Aug. 2023*
- The Best Presentation Award at Master Thesis Midterm Presentation, Tokyo Institute of Technology | *Jul. 2020*
- NMR Society of Japan Young Scientists Poster Award | *Nov. 2020*
- Nakatani Foundation Graduate Fellowship (Granted approx. 35,000 USD) | *Sept. 2019 – Aug. 2021*
- Nakatani RIES Fellowship (fully funded research internship at Rice University and Yale University) | *Aug. 2018*

SKILLS

- **Programming**: Python, CUDA, MATLAB, Git, Scripting (Bash), LaTeX, HTML, Vim, Tcl
- **Software**: Machine learning (PyTorch, TensorFlow, Scikit-learn, PyTorch Lightning, WandB, Transformers), Material simulations (ABINIT, Quantum Espresso, VASP, Wannier90, FEFF, Phonopy, Phono3py, OpenMC, Kwant), NMR simulations (SIMPSON, SpinEvolution), Linux, Origin, ChemDraw, Claude code, Codex
- **Material characterization**: Solid-state NMR, Powder X-ray diffraction, Metal-organic framework synthesis

PUBLICATIONS

1. T Matsunaga†, **R Okabe**† T Matsunaga, Y Liu, Y Shigemitsu, Y Ishii, "Design of efficient solvent-suppression scheme in solid-state NMR: Echo-formation mechanism and dephasing by adiabatic inversion pulses," *J. Magn. Reson. 108183*, **2026**.
2. T Isogawa†, **R Okabe**†, N Phadetsuwannukun, M Li, and P Cappellaro*, "Agentic AI for Scientific Reasoning in Autonomous Quantum Sensing Experiments," *arXiv: 2607.25145*, **2026**.
3. A Chotrattanapituk†, **R Okabe**†, E Rha, M Al-Hinai, E Jiang, D Pajerowski, Y Cheng, JJ Turner, and M Li*, "Universal Magnetic Structure Prediction from Atomic Coordinates with Near-Experimental Accuracy," *arXiv:2605.16230*, **2026**.
4. MJ Landry†, **R Okabe**†, C Fu, M Li, "Quantum Theory of Functionally Graded Materials," *arXiv:2603.03424*, **2026**.
5. M Cheng†, C Fu†, **R Okabe**†, A Chotrattanapituk†, A Boonkird, NT Hung and M Li, "Artificial intelligence-driven approaches for materials design and discovery," *Nat. Mater. 25, 174–190*, **2026**.
6. M Mandal†, E Rha†, A Chotrattanapituk†, D Cordova Carrizales†, A Lygo, KB Woller, M Cheng, **R Okabe**, G Zhu , S Kishida, K Mak, CL Fu, S Michalopoulos, C Liu, L Wu, Y Zhu, S Stemmer, M Li, "Tuning Chiral Anomaly Signature in a Dirac Semimetal via Fast-Ion Implantation," *Nano Lett. 25, 51, 17571–17577*, **2025**.
7. A Boonkird†, M Cheng†, A Chotrattanapituk†, D Cordova Carrizales, **R Okabe**, NC Drucker, M Mandal, T Nguyen, J Yeo, V Belosevich, E Spero, C Ortiz, Q Ma, L Fu, T Palacios, F Niroui and M Li, "Are Quantum Materials Economically and Environmentally Sustainable?", *Materials Today*, **2025**.
   - Press release at MIT News.
8. **R Okabe**, M Cheng, A Chotrattanapituk, M Mandal, K Mak, D Cordova Carrizales, NT Hung, X Fu, B Han, Y Wang, W Xie, RJ Cava, TS Jaakkola, Y Cheng, M Li, " Structural Constraint Integration in a Generative Model for the Discovery of Quantum Materials," *Nat. Mater.* **2025**.
   - Press release at MIT News.
9. C Fu†, M Cheng†, NT Hung†, E Rha, Z Chen, **R Okabe**, D Cordova Carrizales, M Mandal, YQ Cheng and M Li, "AI-driven defect engineering in advanced thermoelectric materials," *Adv. Mater. 37, 2505642*, **2025**.
10. M Cheng, R Okabe, and M Li, "Closing the superconducting gap with AI," *Newton 1, 100093*, **2025**.
11. B Han†, **R Okabe**†, A Chotrattanapituk†, M Cheng†, M Li and Y Cheng, "AI-Powered Exploration of Molecular Vibrations, Phonons and Spectroscopy." *Digit. Discov.*, 4(3), 584-624, **2025**.
12. **R Okabe**†, Z West†, A Chotrattanapituk, M Cheng, D Cordova Carrizales, W Xie, RJ Cava and M Li, "Large Language Model-Guided Prediction Toward Quantum Materials Synthesis," *arXiv:2410.20976*, **2024**.
13. P Siriviboon†, C Fu†, M Landry†, **R Okabe**, D Cordova Carrizales, Y Wang and M Li, "Quantum Theory of X-ray Photon Correlation Spectroscopy," arXiv:2412.03635, **2024**.
14. NT Hung, **R Okabe**, A Chotrattanapituk and M Li, "Universal Ensemble-Embedding Graph Neural Network for Direct Prediction of Optical Spectra from Crystal Structures," *Adv. Mater., 2409175*, **2024**.
   - Press release at Tohoku University Research News, Graphical Abstract.
15. **R Okabe**†, A Chotrattanapituk†, A Boonkird, N Andrejevic, X Fu, TS Jaakkola, Q Song, T Nguyen, NC Drucker, S Mu, B Liao, Y Cheng, M Li, "Virtual Node Graph Neural Network for Full Phonon Prediction," *Nat Comput Sci 4, 522–531*, **2024**.
   - Journal Research Briefing, July 2024 issue Front Cover, press release at MIT News.
16. M Mandal†, A Chotrattanapituk†, K Woller, H Xu, N Mao, **R Okabe**, A Boonkird, T Nguyen, NC Drucker, T Momiki, J Li, J Kong, and M Li, "Precise Fermi-level engineering in a topological Weyl semimetal via fast ion implantation," *Appl. Phys. Rev. 11, 021429*, **2024**.
   - Press release at MIT News.
17. M Cheng, **R Okabe**, A Chotrattanapituk, M Li, "Machine Learning Detection of Majorana Zero Modes from Zero Bias Peak Measurements," *Matter 7, 2507*, **2024**.
18. **R Okabe**†, S Xue†, J Vavrek†, J Yu, R Pavlovsky, V Negut, B Quiter, J Cates, T Liu, B Forget, S Jegelka, G Kohse, L-W Hu, M Li, "Tetris-Inspired Detector with Neural Network for Radiation Mapping", *Nat. Commun. 15, 3061*, **2024**.
   - Press release at MIT News, Physics World, Nuclear Engineering International.
19. NC Drucker, T Nguyen, M Mandal, P Siriviboon, Y Quan, A Boonkird, **R Okabe**, F Li, K Burrage, F Funuma, M Matsuda, D Abernathy, T Williams, S Chi, F Ye, C Nelson, B Liao, P Volkov, and M Li, "Incipient Nematicity from Electron Flat Bands in a Kagome Metal", *arXiv:2401.17141*, **2024**

20. **R Okabe**, M Li, Y Iwasaki, N Regnault, C Felser, M Shirai, A Kovacs, T Schrefl, A Hirohata. "Materials Informatics for the Development and Discovery of Future Magnetic Materials", *IEEE Magentic Letters*, 14, *1-5*, **2023**.
    - IEEE Magnetics Letters (Volume: 14) Front Cover.
21. M Mandal, NC Drucker, P Siriviboon, T Nguyen, A Boonkird, TN Lamichhane, **R Okabe**, A Chotrattanapituk and M Li, "Topological superconductors from a materials perspective", *Chem. Mater.* 35, 16, 6184–6200, **2023**.
22. NC Drucker, T Liu, Z Chen, **R Okabe**, A Chotrattanapituk, T Nguyen, Y Wang, M Li, "Challenges and Opportunities of Machine Learning on Neutron and X-ray Scattering", *Synchrotron Radiat. News*, *35, 16-20*, **2022**.
23. HA Merker, H Heiberger, L Nguyen, T Liu, Z Chen, N Andrejevic, NC Drucker, **R Okabe**, Y Wang, T Smidt, M Li, "Machine Learning Magnetism Classifiers from Atomic Coordinates" *iScience*, *25, 10, 105192*, **2022**.
24. T Matsunaga, **R Okabe**, Y Ishii, "Efficient solvent suppression with adiabatic inversion for ^{1}H-detected solid-state NMR" *J. Biomol. NMR. 12, 365-370, 2024*, **2021**.
25. B Pattengale, JG Freeze, M Guberman-Pfeffer, **R Okabe**, S Ostresh, S Chaudhuri, VS Batista, CA Schmuttenmaer, "A Conductive Metal-Organic Framework Photoanode," *Chem. Sci. 11*, *9593–9603*, **2020**.
    - Conference abstract proceedings (2022).
26. B Pattengale, J Neu, S Ostresh, G Hu, JA. Spies, **R Okabe**, GW Brudvig, CA Schmuttenmaer. "Metal-Organic Framework Photoconductivity *via* Time-Resolved Terahertz Spectroscopy," *J. Am. Chem. Soc. 141*, 9793–9797, **2019**.

PRESENTATIONS

1. **Innovation Spotlight: Japan**, *Boston, MA, USA | May 2026*

Invited exhibition at the STEM showcase presented by the Consul General of Japan in Boston and the Museum of Science [website]

2. **UMass Boston Physics Colloquium**, *Boston, MA, USA | Mar 2026*

Invited Talk: "Accelerating Quantum Material Discovery with Machine Learning: From Property Prediction to Materials Design"

3. **APS Global Physics Summit 2026**, *Denver, CO, USA | Mar. 2026*

Tutorial 4: "Generative AI for Physics: From Models to Materials" [Session website] [Colab demo]

Oral: "Structural Constraint Integration in a Generative Model for the Discovery of Quantum Materials"

4. **Laboratory of Atomic and Solid State Physics (LASSP) Quantum Theory Seminar,** *Ithaca, NY, USA | Oct. 2025*

Invited talk: "Advancing Quantum Material Discovery with Geometric Machine learning"

5. **IMRC 2025 MRS/SMM Student Poster Award Exchange Program,** *Cancún, Mexico | Aug. 2025*

Invited poster: "Structural Constraint Integration in Generative Model for Discovery of Quantum Material Candidates"

6. **FT Workshop AI-Accelerated Superconductor Discovery,** *Gainesville, FL, USA | April. 2025*

Poster: "Structural Constraint Integration in Generative Model for Discovery of Quantum Material Candidates"

7. **2025 MRS Spring Meeting,** *Seattle, WA, USA | Apr. 2025*

Award Talk & Poster: "Advancing Quantum Material Discovery through Structural Constraint Integration in Generative Models"

8. **Competing Orders in Quantum Materials,** *New Haven, CT, USA | Mar 2024*

Oral: "Advancing Quantum Material Discovery through Structural Constraint Integration in Generative Models"

9. **2024 MRS Fall Meeting,** *Boston, MA, USA | Dec. 2024*

Oral: "Structural Constraint Integration in Generative Model for Discovery of Quantum Material Candidates"

Oral: "Large Language Model-Guided Prediction Toward Quantum Materials Synthesis"

10. **Northeast Quantum Forum 2024,** *Durham, NH, USA | Oct 2024*

Invited Talk: "Machine-learning assisted study of quantum material properties and quantum material discovery"

11. **Quantum Materials for Emergent Applications in Quantum Science**, *Telluride, CO, USA | June 2024*

Invited Talk: "Structural Constraint Integration in Generative Model for Discovery of Quantum Material Candidates"

12. **2024 ACS Spring Meeting** *New Orleans, LA, USA | Mar. 2024*

Poster: "Virtual Node Augmented Machine Learning for Material Property Prediction"

13. **2023 MRS Spring Meeting** *San Francisco, CA, USA | Apr. 2023*

Poster: "Virtual Node Augmented Machine Learning for Material Property Prediction"

14. **2022 MRS Fall Meeting** *Boston, MA, USA | Nov-Dec. 2022*

Poster: "Machine Learning Gamma Phonons from Atomic Coordinates"

15. **2022 MRS Spring Meeting** *(online) | May. 2022*

Oral: “Application of Radiation Detection Materials for Radiation Mapping with Machine Learning”

16. **22nd International Society of Magnetic Resonance Conference** *(online) | Aug. 2021*

Poster: “Theoretical explanation of new solvent suppression scheme with adiabatic pulse and application for solid-state NMR experiments”

17. **The 59th Annual Meeting of the NMR Society of Japan**, *Gunma, Japan | Nov. 2020*

Poster Presentation: “Development of Suppression Pulse with Adiabatic Pulse and Application for Biological Solid-State NMR”

18. **The 58th Annual Meeting of the NMR Society of Japan**, *Kanagawa, Japan | Nov. 2019*

Poster Presentation: “Development of Suppression Pulse with Adiabatic Pulse for Solid-State NMR Experiment”

COMMUNITY SERVICE

- Journal Peer Review: Matter., *NPJ Quantum Mater., Phys. Rev. L., Phys. Rev. M., Phys. Rev. A., EPJ Quantum Technol., 2026 Conference on Physics and AI (PAI26) @Stanford*
- Session chair: APS Global Physics Summit 2026 (MAR-J42, MAR-W27), PAI26